SECTION 213

TIMBER BRIDGES – CONSTRUCTION

213.01.01 General – Timber materials may be used in the completed Work only when explicitly identified in the Contract as an acceptable product.

213.02 Materials, Handling, Storage and Construction – All materials, handling, storage and construction shall be as specified and shall conform to the relevant requirements of SS 214, SS 904, SS 905, SS 906, and SS 908.

All materials shall be handled and stored with reasonable care to prevent damage such as puncture, cutting or crushing of fibre.

213.03 Logs for Cribs and Log Stringers – Logs for cribs shall be cut from live trees and shall be peeled before use.

Log stringers shall be cut from live trees not less than 30 days before use and shall be seasoned with the bark on. Immediately before use, all bark shall be peeled and the log trimmed smooth of all knots and projections.

213.04 Work Practice – The standard of work shall be first-class throughout and in accordance with AASHTO LRFD Bridge Construction Specifications.

Details for construction of log stringer bridges and log cribs shall be in accordance with "Log Bridge Construction Handbook 1980" published by Forest Engineering Research Institute of Canada (FERIC), except as otherwise specified in the Contract.

213.05 Framing – All the cutting, boring, framing, match marking, etc. required on all timber shall be done by competent bridge framers in a thorough manner, in accordance with good work practice. It shall be done such that surfaces in contact shall bear evenly and fully; no shims or open joints are permitted. All measurements shall be accurate. Gains and daps shall have plane, smooth surfaces.

Holes shall be bored with an auger to the sizes shown in Table 213-A.

Table 213-A: Hole Sizes

COMPONENT	FIR AND HARD-WOODS	CEDAR AND SOFT-WOODS
BOLTS	1/16" Large	Same size as bolt
DRIFT BOLTS	Same size as bolt	1/16" Small
DOWELS	1/16" Large for ends of posts, but in sills make same size as dowel	1/16" Large for ends of posts, but in sills make same size as dowel

Spikes that are likely to crack or unnecessarily damage the timber shall have holes bored 1/16" small.

Hardware and metalwork in contact with treated wood shall be galvanized or stainless steel and shall be in accordance with the requirements of SS 908.08.

213.06 Field Treatment – Cutting, framing and boring of timbers to receive preservative treatment shall be done before treatment insofar as possible.

In the event that cutting or drilling becomes necessary after treatment, a field treatment preservative specified in CSA O80.3 or the American Wood Protection Associated (AWPA) Standard M4 shall be used and applied in accordance to its label. At least two coats shall be applied and where possible, the colour of the preservative treatment used for protecting field cuts shall match the original preservative treatment colour.

For bridge components, creosote and copper naphthenate shall be the only permitted field preservatives.

For untreated timber, all surfaces in contact with any other surface and all bolt holes shall be swabbed with a water-borne preservative.

Preservative chemicals shall not be permitted to spill onto the ground or into any waterway during field treatment.

213.07 Preparation of Laminated Decks for Wearing Surface

213.07.01 For Asphalt Plank – Depressions in laminated decks shall be levelled by placing and compacting a fine sand levelling course.

The levelling course shall consist of a mixture of fine durable sand conforming to the gradation shown in Table 213-B.

The sand shall be thoroughly mixed with asphalt emulsion, Type SS-1 which shall conform to the requirements of SS 952.08. The asphalt emulsion content shall be such that an air void content of 5% to 8% in the compacted mixture is obtained. This will normally require an asphalt emulsion of 15% to 25% by weight of the total mixture.

Before laying and nailing asphalt plank, laminated decks shall be swabbed with oxidized mopping asphalt Type 2 conforming to the requirements of Table 213-C.

213.07.02 For Asphalt – Prior to placing asphalt wearing surface, creosote-treated laminated decks shall be lime-scrubbed where creosote oil has "bled" to the surface.

To prevent "bleeding" of the creosote oil, the mix temperature of hot-mix asphalt placed on these decks shall be as low as is practicable.

213.08 Basis of Payment – Payment will be made at the unit prices bid, for the various items in the Schedule of Approximate Quantities and Unit Prices, for material remaining in and forming a part of the finished structure. Such payment shall be accepted as full compensation for all labour, equipment, tools, materials, including hardware and furnishing and placing all other items required to complete the work.

Table 213-B: Consistency of the Levelling Course

STANDARD SIEVE SIZES		PERCENTAGE PASSING BY WEIGHT (%)
ASTM	METRIC	
#8	2.36 mm	100
#16	1.18 mm	60 – 100
#30	0.600 mm	46 – 80
#50	0.300 mm	35 – 62
#100	0.150 mm	25 – 46
#200	0.075 mm	18 – 35

Table 213-C: Requirements for Oxidized Mopping Asphalt Type 2

TEST DESCRIPTION	ASTM TEST METHOD	TYPE 2 ASPHALT	
		MINIMUM	MAXIMUM
Softening Point (R & B), °C	D36	74	79
Flash Point, C.O.C., °C	D96	232	---
Penetration @ 0°C	D5	12	20
Penetration @ 25°C	D5	20	35
Penetration @ 46°C	D5	---	70
A.P.I. Gravity @ 15°C	D71	4.6	5.8
Ductility @ 25°C, 5 cm minimum	D113	3	---
Loss on Heating to 163°C (325°F), 5 hr., %	D6	---	0.3
Penetration of Residue, % of original	D5	80	0
Total Bitumen (soluble in CS_2) %: Mineral Stabilized Asphalt	D4	80	---
Total Bitumen (soluble in CS_2) %: Asphalt Without Mineral Stabilizer	D4	99	---

SECTION 214

TIMBER PILING CONSTRUCTION

Note: Timber piles may be used in the completed Work only when explicitly identified in the Contract as an acceptable material.

214.01 Materials – All piling shall conform to requirements of SS 906 and preservative treated piles shall meet the requirements of SS 908.

Hardware, fasteners and metalwork for timber piling construction shall be galvanized or stainless steel and shall also meet the requirements of SS 908.08.

214.02 Pile Lengths – Piling shall be supplied to the minimum lengths specified. At the Contractor's expense, allowances will be made for such additional lengths of pile as may be required to provide fresh heading and to suit the Contractor's methods of operation.

Pile lengths determined by the Ministry, are those expected to remain in the completed structure plus 600 mm for cut-off, except for end bearing piles where the allowance for cut-off and bearing depth irregularity is 3.5 m or as otherwise specified on the Drawings or the Special Provisions.

214.03 Handling of Piles – Handling of piles shall be in accordance with SS 906.08 and SS 908.

214.04 Hammer – Piles may be driven with a drop hammer or a mechanical hammer. Drop hammers shall weigh not less than 1,400 kg for piles less than 15 m long and not less than 1,600 kg for piles 15 m and over in length, and in no case shall the weight of the hammer be less than the combined weight of driving head and pile. Mechanical hammers shall develop a total energy not less than 3,900 kg m per blow.

214.05 Leads – Pile drivers with fixed leads or with hanging or swinging leads that can be held in a fixed position during driving shall be used to drive all piles.

Battered piles shall be driven using leads adaptable to their driving.

When driving treated timber piles, the use of spuds and chocks in the leads shall be kept to a minimum in order that the treated surface is not damaged.

214.06 Preparation for Driving

214.06.01 Driving Caps and Bands – The heads of timber piles, shall be protected by suitable driving caps of approved design, preferably having a rope or other suitable cushion next to the pile head and fitted into a casting which in turn supports a timber shock block. Alternatively, the pile head may be banded with a pile ring and covered with a steel wire mat. When driving treated piles, a driving cap shall be used.

214.06.02 Pointing – Steel points shall be supplied and installed as shown on the Drawings or Special Provisions.

Where soil conditions, in the opinion of the Ministry Representative, require the pointing of piles, the Contractor shall supply and install the piles with steel points fabricated in accordance with SS Drawing SP214-01 Pile Shoe Type A.

214.06.03 Excavation – Footing excavation and seal excavation, where applicable, shall be completed with any necessary allowance made for upheaval before the driving of footing piles. Should too great an allowance be made, backfilling with gravel will, in general, be permitted to raise the bottom of the excavation to the correct elevation. Any material forced up between the piles to above the footing level shall be removed to the correct elevation. No extra payment will be made for the supply and placing of gravel or any additional excavation to obtain the correct elevation.

214.07 Driving – Unless otherwise specified, all piles shall be driven with the small end downward. The fall of a drop hammer shall be regulated to avoid damage to the piles and under no circumstances shall the drop exceed 3 m.

Each pile shall be square-headed at the top, properly pointed at the tip, and set properly in the leads; if any pile becomes split or broomed on the top it shall be fresh headed.

Piles shall be driven to the minimum depth (maximum tip elevation) indicated on the Drawings or as otherwise specified and unless ordered otherwise by the Ministry Representative, to practical refusal, which shall be to the depth at which the piles have an allowable load resistance of 20 tonnes and an average penetration per blow determined by the following Engineering News Record (ENR) formulae:

Equation 214-1: For Drop Hammers

$$s = \frac{wh}{6p} - 0.03$$

Equation 214-2: For Mechanical Hammers

$$s = \frac{wh}{6p} - 0.003$$

Where:

s = average penetration per blow in metres under last 10 blows
w = weight of moving part of hammer in newtons
h = effective fall of hammer in metres
p = allowable load of pile in newtons

Where it is required by the Ministry Representative to penetrate a thin stratum (which may show practical refusal in driving) to obtain greater depth in a less resistant formation, driving shall be continued through the stratum regardless of the definition laid down for practical refusal.

Any pile which, in the opinion of the Ministry Representative, is damaged by driving so as to be deemed unfit for the use for which it is intended, or is so far out of position, or off vertical or designated batter, as to require excessive force or manipulation to restore it to its proper position, shall be withdrawn and replaced by a new pile. If this is not possible, a new pile shall be driven adjacent to the defective pile. The damaged, out-of-position or misaligned piles shall be for the Contractor's account.

Where boulders or other large obstructions make it impossible to drive piles in the locations shown on the Drawings, the Ministry Representative may direct the Contractor to drive an additional pile, or piles, for which the Contractor will be paid in accordance with the various Items of SS 214.11 – Payment.

214.08 Additional Equipment – Where, in the opinion of the Ministry Representative, the bounce of the hammer is excessive or where the required depth or resistance is not obtained by the use of a hammer complying with the above minimum requirements the Contractor shall provide a heavier hammer at the Contractor's expense.

214.09 Accuracy of Driving – All piles shall be driven in such a manner that any variation from the vertical, or batter as shown on the Drawings, shall not exceed 20 mm/m. Piles in trestle bents shall be driven so that the cap may be placed in its proper location without incurring excessive stresses in the piles. Cutting, springing, dapping or facing-off for placing bracings and planking on piles will not be permitted.

After driving, footing piles shall not be more than 75 mm out of the horizontal position shown on the Drawings.

214.10 Pile Heads

214.10.01 Cut-off – The tops of all piling shall be cut off to a true plane as shown on the Drawings or as established by the Ministry Representative. Piles shall show a solid head at the plane of cutoff.

214.10.02 Cut-off Treatment – Pile heads not encased in concrete shall be protected as follows.

(a) Treated Piles – After cut-off, the pile heads shall be covered with alternate layers of hot roofing asphalt and loosely woven glass fabric, using four applications of asphalt and three layers of fabric.

The fabric shall measure at least 150 mm more in dimension than the diameter of the pile and shall be neatly folded down and secured by large headed galvanized roofing nails or by binding with not less than seven complete turns of galvanized wire securely held in place by large-headed galvanized roofing nails or staples. The edges of the fabric projecting below the fastening shall be neatly trimmed.

(b) Untreated Piles – After cut-off, the pile head shall be thoroughly swabbed with three coats of hot creosote oil and if they are to be capped, they shall also be coated with hot roofing asphalt.

214.11 Payment – Payment for timber piling construction will be made under the following Items.

214.11.01 Mobilization and Fixed Costs for Pile Installation – Payment will be made at the Lump Sum price bid for providing, setting up and removing the plant and equipment for driving the piles, and shall be accepted as full compensation for everything furnished and done in connection therewith.

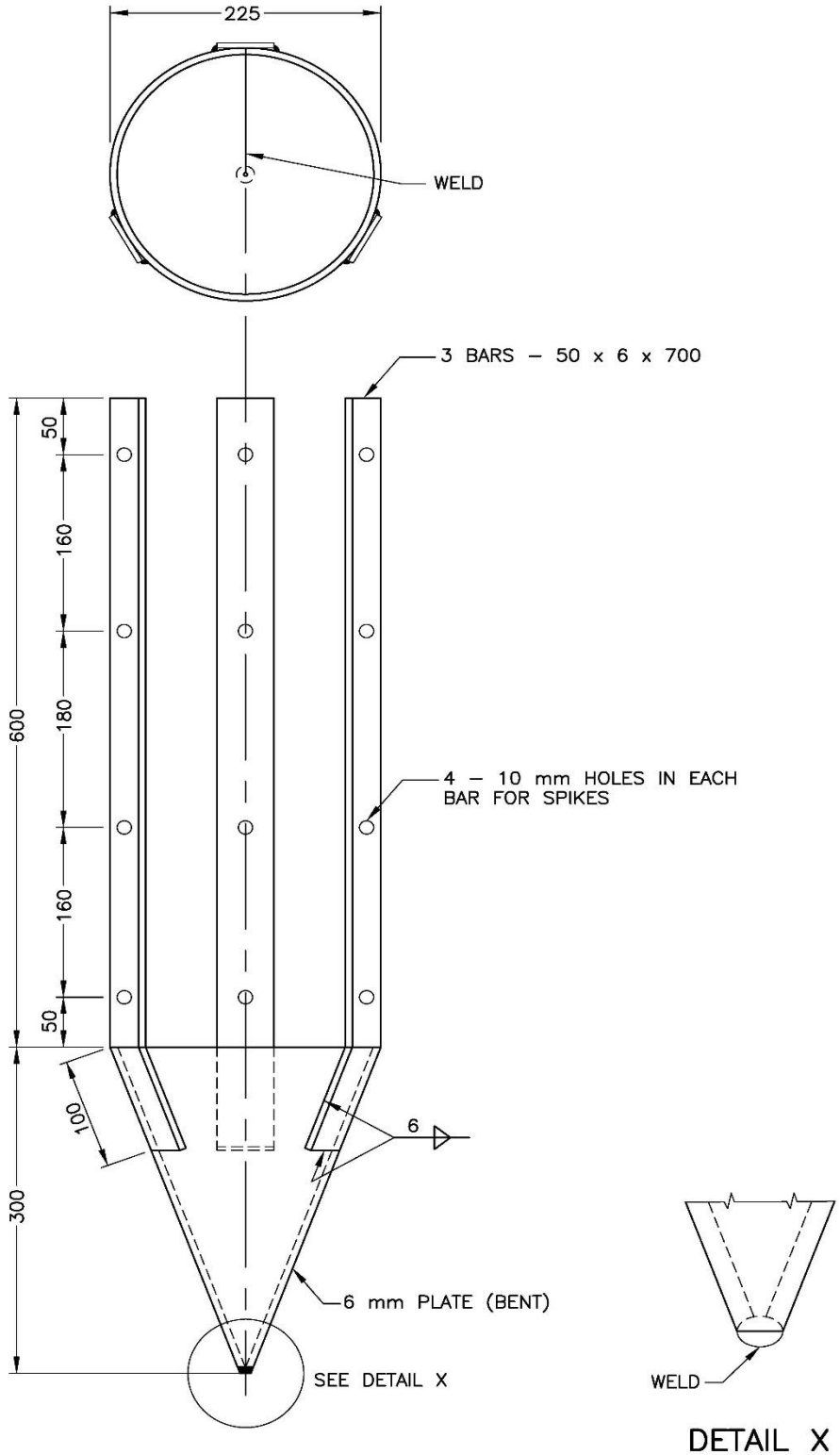
214.11.02 Supply – Payment will be made at the Unit Price bid per metre for the total number of metres of piling as shown on the Drawings or as specified by the Ministry Representative. Such payment shall be accepted as full compensation for everything furnished and done in connection with the supply and delivery of the piles.

214.11.03 Installation – Payment shall be accepted as full compensation for all costs in connection with supply and installation of pile points and installation of piles.

If the length of piling is more than anticipated, the Ministry will pay documented costs of delivery of less than full loads of additional piling, without markup.

If pile points are not specified on the Drawings or Special Provisions, but are ordered by the Ministry Representative, payment will be as Extra Work.

PILE SHOE TYPE A	SP214–01



NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

SECTION 215
BRIDGES

215.01 General - Bridges shall be designed in accordance with the Canadian Highway Bridge Design Code (CHBDC), CSA S6 and the Ministry's Bridge Standards and Procedures Manual, including the Ministry's Supplement to CHBDC S6.

Construction shall be in accordance with the current Ministry standards, practices and the Contract documents prepared for the particular structure.

SECTION 216
COATING OF STEELWORK

216.01 Scope – This Section covers the operations required for shop, field and/or maintenance coating of structural and miscellaneous steel surfaces. The coating process includes pre-cleaning, surface preparation, coating, and curing of coatings.

216.02 Definitions

(a) Good Painting Practice – The application of generally accepted methods and equipment to clean, contain and coat industrial structures in a safe, practical, environmentally sensitive manner resulting in a long-lasting quality protective coating system. These principals are laid out in this Section. Many of the points were drawn from the SSPC (Society for Protective Coatings) publication Steel Structures Painting Manual Volume 1, Good Painting Practice. References are to the current edition unless otherwise stated. These practices are based on experience in the United States and where they conflict with Canadian or British Columbia laws, regulations or guidelines, the local regulations shall prevail.

(b) Shop, Field and Maintenance Coating – This consists of the series of operations that include the surface preparation, pre-treatment (if required) and application of coatings to structural steel surfaces either in the shop or in field. It also includes the supplying of all labour, equipment and materials. In addition, it encompasses the protection of newly coated surfaces during curing, transport and erection as well as the protection of traffic, property and the environment. Additionally, it also includes the disposal of waste material in accordance with all applicable environmental, fisheries and local regulations and bylaws.

(c) Painting / Coating – The terms painting and coating are considered synonymous for the purposes of this document. Techniques such as metallizing or galvanizing, while not typically thought of as painting, are considered here to be painting or coating since they form part or all of a protective system.

(d) Shop Coating – Shop coating is the coating of steel or other components in a shop or plant before shipment to the site of erection. This may include part of a coating system or the whole system including the topcoat.

(e) Field Coating – Field coating is the coating of steel or components at the place of erection.

(f) Maintenance Coating – Maintenance coating is the coating of structures in service that have previously been partially or completely coated. The maintenance coating may be one of four general types:

(i) Touchup – Touchup is the cleaning and coating of small selected areas on a structure. Typically, this is used to repair damaged or failed areas of coating. For repairs, sufficient coats are added to bring the new coating up to the level of the existing coating surrounding the repairs. It may also be used as an interim maintenance strategy to extend the lifetime of a coating system overall.

(ii) Zone Coating – Zone coating is the coating process where specific whole parts of a structure are coated and others are deliberately left. This deals with larger portions of a structure than touchup coating. For example, an above deck arch portion of a bridge would be zone coated, but the edges of the flanges would be touched up. Zone coating may use touchup, overcoat, or recoat strategies to meet its objectives. Zone coating also includes coating new steel added to old structures for example steel added for seismic upgrades.

(iii) Overcoating – Overcoating is the coating of portions of a structure where the structure is touched up, then coated with a topcoat overall. It may include top coating part or all of the structure.

(iv) Recoating – Recoating is the coating process where larger portions of old coating on a structure are removed to bare metal and a completely new coating system is applied over the entire area.

(g) Anchor Profile – The anchor profile is a measure of the roughness of the prepared surface. The profile is normally measured from surface peak to surface valley and expressed as a range (for example 50 to 75 µm or 2 to 3 mils). Besides the depth, the surface profile is dependent on the grit type used to create it. Mineral abrasives such as sand, slag, garnet, etc. produce a 'sharp' profile. Steel shot produces a rounded profile unless mixed with 15% steel grit. Normally the anchor profile is specified for both depth and sharpness.

(h) Containment / Hoarding – The process of encapsulating the work site so that hazardous or nuisance materials cannot escape to the surrounding environment. SSPC Guide 6 Guide for Containing Debris Generated During Paint Removal Operations specifies types and levels of containment necessary for various cleaning operations.

(i) Ambient Conditions – Ambient conditions refer to such things as wind, relative humidity, dew point, air temperature, substrate temperature etc. that affect the cleaning, application, and curing of coatings. They are very important in obtaining a properly adhered, long life coating. All coating works shall be within the specified parameters as given in SS 216.08.04, SS 216.08.05 and SS 216.08.09. Where ambient environmental conditions do not fall within these conditions, coating work may continue if the conditions are being met within the containment. This may mean extra hoarding, dehumidification, ventilation and/or temperature control for the duration of cleaning, application and curing of the coating system.

It is worthy of note that some coatings are designed to be applied in conditions outside those mentioned above. These coatings and their conditions of use will be subject to the approval of the Ministry Representative prior to use.

216.03 Reference Standards – It is recognized that standards are changed or amended from time to time. All specifications mentioned shall be the current editions at the date and time of tendering unless a specific edition of a reference or standard is stated here or in the Contract Drawings and Special Provisions.

216.03.01 Ministry of Transportation and Infrastructure

(a) Recognized Products List (RPL) – The Ministry publishes a Recognized Products List indicating products that are considered generally acceptable for particular purposes. This is available on the Ministry website at:

https://www2.gov.bc.ca/assets/gov/driving-and-transportation/transportation-infrastructure/engineering-standards-and-guidelines/recognized-products-list/recognized_products_list.pdf?forcedownload=true

Only those coating systems and products listed under the heading of "Coating Systems" in the RPL or as otherwise specified in the Special Provisions or on the Drawings shall be used on Ministry Projects. The Ministry may, at its sole discretion, designate alternative materials to fit a unique situation.

216.03.02 AMPP – Association for Materials Protection and Performance – A professional association formed by the merger of NACE International and SSPC in January 2021. Specifications continue to be issued under the NACE and SSPC nomenclature.

(a) NACE – Was the "National Association of Corrosion Engineers", publishing specifications and test procedures for coatings and corrosion engineering.

(b) SSPC – Was the "Society for Protective Coatings" AAMP/SSPC publishes two volumes: SSPC Painting Manual Volume 1, Good Painting Practice and SSPC Painting Manual Volume 2, Systems and Specifications, which lay out the basics for painting steel structures.

The following AAMP / SSPC / NACE standards and guidelines shall apply unless noted otherwise.

(c) Surface Preparation Standards

(i) General – SSPC-SP COM Surface Preparation Commentary for Steel and Concrete Substrates.

(ii) Visual Standards:

(A) SSPC-VIS 1, Guide and Reference Photographs for Steel Surfaces Prepared by Dry Abrasive Blast Cleaning

(B) SSPC-VIS 2 Standard Method of Evaluating Degree of Rusting on Painted Steel Surfaces

(C) SSPC-VIS 3 Guide and Reference Photographs for Steel Surfaces Prepared by Hand and Power Tool Cleaning

(D) SSPC-VIS 4/NACE VIS 7 Guide and Reference Photographs for Steel Prepared by Water Jetting

(E) SSPC-VIS 5/NACE VIS 9 Guide and Reference Photographs for Steel Surfaces Prepared by Wet Abrasive Blast Cleaning

(iii) Surface Preparation Standards:

(A) SSPC-SP 1 Solvent Cleaning

(B) SSPC-SP 2 Hand Tool Cleaning

(C) SSPC-SP 3 Power Tool Cleaning

(D) SSPC-SP 5/NACE No. 1 White Metal Blast Cleaning

(E) SSPC-SP 5 (WAB)/NACE WAB-1, White Metal Wet Blast Cleaning

(F) SSPC-SP 6/NACE No. 3 Commercial Blast Cleaning

(G) SSPC-SP 6 (WAB) / NACE WAB-3 Commercial Wet Blast Cleaning

(H) SSPC-SP 7/NACE No. 4 Brush-off Blast Cleaning

(I) SSPC-SP 7 (WAB)/NACE WAB-4 Brush-Off Wet Blast Cleaning

(J) SSPC-SP 10/NACE No. 2 Near-White Blast Cleaning

(K) SSPC-SP 10 (WAB)/NACE WAB-2 Near White Wet Blast Cleaning

(L) SSPC-SP 11 Power Tool Cleaning to Bare Metal

(M) SSPC-SP 13/NACE No. 6 Surface Preparation of Concrete

(N) SSPC-SP 14/ NACE No. 8 Industrial Blast Cleaning

(O) SSPC-SP 14 (WAB)_/_NACE WAB-8 Industrial Wet Blast Cleaning

(P) SSPC-SP 15 Commercial Grade Power Tool Cleaning

(Q) SSPC–SP 16 Brush-off Blast Cleaning of Non-Ferrous Metals

(R) SSPC-SP WJ-1/NACE WJ-1 Waterjet Cleaning of metals Clean to Bare Substrate

(S) SSPC-SP WJ-2/NACE WJ-2 Waterjet Cleaning of metals Very Thorough Cleaning

(T) SSPC-SP WJ-3/NACE WJ-3 Waterjet Cleaning of metals Thorough Cleaning

(U) SSPC-SP WJ-4/NACE WJ-4 Waterjet Cleaning of metals Light Cleaning

(iv) Abrasive Specifications:

(A) SSPC-AB 1 Mineral and Slag Abrasives

(B) SSPC-AB 2 Cleanliness of Recycled Ferrous Metallic Abrasives

(C) SSPC-AB 3 Ferrous Metallic Abrasive

(D) SSPC-AB 4 Recyclable Encapsulated Abrasive Media (in a compressible cellular matrix)

(v) Application Standards, Guides and Specifications:

(A) SSPC-PA COM Commentary on Paint Application

(B) SSPC-PA 1 Shop, Field and Maintenance Painting of Steel

(C) SSPC-PA 2 Procedure for Determining Compliance to Required DFT, Restriction Level 2

(D) SSPC-PA 17, Procedure for Determining Conformance to Steel Profile/Surface

(vi) Guides:

(A) SSPC – Guide 6 Guide for Containing Surface Preparation Debris

(B) SSPC – PA Guide 11 Protecting Edges, Crevices, and Irregular Steel Surfaced by Stripe Coating

(C) SSPC – Guide 12 Guide for Illumination of Industrial Painting

(D) SSPC – Guide 14 Guide for the Repair of Imperfections in Galvanized, Organic or Inorganic Zinc-Coated Steel Using Organic Zinc-Rich Coatings

(E) SSPC – Guide 15 Field Methods for Extraction and Analysis of Soluble Salts on Steel and Other Nonporous Substrates

(vii) Technical Updates:

(A) SSPC-TU 3 Overcoating

(B) SSPC-TU 6 Chemical Stripping of Organic Coatings from Steel Surfaces

(C) SSPC-TU 7 Conducting Ambient Air, Soil, and Water Sampling During Surface Preparation and Paint Disturbance Activities

(viii) Qualification Procedures – These requirements will be laid out specifically in the Contract for each project and will depend primarily on the work to be done. Contractors may require current certification in one or more of the following levels to perform Ministry work as called for in the particular tender.

(A) SSPC-QP COM Commentary on Qualification Procedures

(B) SSPC-QP 1 Certification (Field Application to Complex Industrial and Marine Structures)

(C) SSPC-QP 2 Certification (Field Removal of Hazardous Coatings)

(D) SSPC-QP 3 Certification (Shop Painting Certification Program)

(E) SSPC-QP 6 Thermal Spray Contractor Qualification

(F) SSPC QP 7 Certification (Painting Contractor Introductory Program)

(G) NACE Coating Inspection Program Level 2

(H) SSPC Bridge Coatings Inspection Program (BCI) Level 2

216.03.03 Hot Dip Galvanizing – ASTM A123 Standard Specification for Zinc (Hot Dip Galvanized) Coatings on Iron and Steel Products.

216.03.04 Metallizing – SSPC-CS23.00 / AWS C2.23M / NACE No. 12, Specification for the Application of Thermal Spray Coatings (Metallizing) of Aluminum, Zinc, and Their Alloys and Composites for the Corrosion Protection of Steel.

216.03.05 Colour Standards – Refer to AMS-STD-595A: Colors Used In Government Procurement. Topcoat colours shall be in accordance with the Special Provisions and/or the Drawings or as determined by the Ministry Representative. For weathering steel, unless noted otherwise, the topcoat colour shall match the expected colour of the oxidized surfaces (Federal Color 30045 unless noted otherwise). The proposed colour shall be subject to the acceptance of the Ministry Representative.

216.04 Environmental Considerations – All coating works including preparation shall comply with SS 165, Protection of the Environment.

216.05 Containment of Coating Operations

216.05.01 General – Full containment shall be provided to prevent contamination of the environment. This enables work to be accomplished by physically limiting the spread of debris and facilitates cleanup. Current containment technology is able to contain virtually all debris generated by washing, water jetting, hand, power tool, or abrasive blasting, and coating operations.

216.05.02 SSPC Guide 6 – Containment shall be provided in accordance with SSPC Guide 6, Guide for Containing Debris Generated During Paint Removal Operations. In general, the Ministry wishes to minimize the environmental impact of all Ministry coating work. To this end, all coating works shall be fully enclosed to prevent contamination of the environment. The coating specification will contain specific language to exactly determine the desired containment end performance requirements. Containment shall be in accordance with the pertinent requirements of the Workers Compensation Act, the Occupational Health and Safety Regulation, and Environment and Fisheries Authorities.

216.05.03 Containment of Debris and Disposal of Debris – All debris from cleaning and coating operations must be contained in such a way as not to contaminate the environment. All debris shall be contained, tested and disposed of in accordance with current environmental, fisheries and local regulations. The Contractor shall provide written proof of disposal of all materials including way-bills and transport logs.

216.06 Construction Site Safety

216.06.01 General – General site safety shall comply with SS 135 Construction Site Safety, the Workers Compensation Act and the Occupational Health and Safety Regulation

216.06.02 Specific Safety Issues Relating to Bridges and Coating Work

(a) Working at Height – All work occurring at height shall conform to all Workers Compensation Act and Occupational Health and Safety Regulation requirements.

(b) Lead Coatings – Many of the Ministry structures have coatings containing lead. While the Ministry will attempt to indicate the presence of lead coatings, it is the responsibility of the Contractor to ascertain the presence and extent of lead coatings if any; and to conform to all regulations for work undertaken. This includes but is not necessarily limited to the Workers Compensation Act, the Occupational Health and Safety Regulation and the Ministry of Environment regulation requirements for working with, storing, transporting, disposing of and documenting any work process involving lead coatings.

(c) Enclosed Spaces – Many of the Ministry structures contain enclosed spaces or spaces that the Workers Compensation Act and the Occupational Health and Safety Regulation define as enclosed spaces. In some instances, the hoarding or containment is considered an enclosed space. It is the responsibility of the Contractor to ascertain whether a space is defined as an enclosed space and to adjust their work procedures to meet all regulation requirements.

216.07 Surface Preparation

216.07.01 General – Surface preparation of areas to be coated shall be in accordance with the specifications herein in this SS 216, the Drawings and the Special Provisions. Where the specified surface preparation differs from the coating manufacturer's recommended surface preparation, the more stringent surface preparation, shall apply, unless approved otherwise by the Ministry Representative.

Where surface preparation is not defined herein this SS 216, the Drawings and the Special Provisions, the manufacturer's specifications shall be used.

216.07.02 Washing

(a) General – Washing is a procedure for removing surface contaminants including oil, grease, drawing and cutting compounds, soil, salts, etc. from either steel or coated surfaces. Removal of the contaminants shall be accomplished using any one or a combination of the following: potable water, alkaline cleaners, solvents, emulsion cleaners, steam cleaning (with detergents), high pressure water, scrubbing (with detergent) or other approved method.

Alkaline cleaners, detergents, solvents, etc. shall not be used over streams, lakes, rivers, etc. or agricultural land without first ensuring compliance with all applicable environmental legislation (federal, provincial and local).

The washing procedure shall be conducted so that any part of the structure which has already been cleaned, or newly coated, shall not be contaminated with the cleaning chemicals or solvents.

Immediately after treatment with cleaning chemicals, the surfaces shall be thoroughly washed with clean water to remove detrimental residues.

(b) Residual Non-Visible Contaminants (NVC) – Immediately prior to abrasive blast cleaning and immediately prior to the application of any coating, surfaces shall be checked for residual non-visible contaminants in accordance with SSPC Guide 15 swabbing extraction method 5.2.5 Latex Sleeve Methodology. Surfaces that exceed any of the following "designated NVC levels" shall be cleaned in accordance with SSPC-SP WJ-4/NACE WJ-4 to achieve these levels:

(i) 7 µg/cm^2 chloride ion,

(ii) 10 µg/cm^2 soluble ferrous ion, and

(iii) 17 µg/cm^2 sulfate

216.07.03 Water Cleaning/Jetting – Waterjet cleaning is the use of clean, pressurized water for removing coatings and other materials, including hazardous materials, from a substrate to achieve a defined degree of surface cleanliness. Waterjet cleaning includes various methods such as low-pressure water cleaning (LP WC), high-pressure water cleaning (HP WC), high-pressure water jetting (HP WJ), and ultra high-pressure water jetting (UHP WJ). Four levels of surface cleanliness are each addressed in separate standards (SSPC-SP WJ-1, to SSPC-SP WJ-4)

Water jetting is inherently dangerous, and safety precautions as listed in SSPC Steel Structures Painting Manual, Volume 1, Good Painting Practice, shall be followed. Where the current Workers Compensation Act and the Occupational Health and Safety Regulation conflict with any of the SSPC-SP WJ specifications, the Workers Compensation Act and the Occupational Health and Safety Regulation shall prevail. Water used in water jetting units shall be clean and free of contaminants that may leave deposits on the surface being cleaned or affect the adhesion of subsequent coatings applied. Water supply to the water jetting units must meet current Federal and Provincial regulations for water taken from streams if on-site water from streams is used.

All waste water and coating debris shall be contained and disposed of in accordance with Environmental and Fisheries Agencies or other applicable regulations.

Wastewater must meet regulations for disposal on-site if it is to be disposed of on-site. These regulations are typically those of Fisheries and Oceans Canada and the Ministry of Environment, but may not be limited to those governmental agencies. As well, SS 165 Protection of the Environment lays out Contractor responsibilities.

216.07.04 Manual and Power Tool Cleaning – Manual cleaning is a procedure for removing loose mill scale, loose rust, loose coating and other detrimental foreign matter by chipping, scraping, grinding, and wire brushing, etc. The tools used for this method of cleaning may be either hand or power tools; they shall be of high quality and maintained in good repair.

The standard of cleaning obtained by this method shall be SSPC-SP 2, SP 3, SP 11, or SP 15 as specified in the Contract. Where surface preparation is not defined in the Contract, the manufacturer's product data sheet specifications shall be used.

In specific circumstances the methods available may be limited by the Drawings and Special Provisions.

In all four preparation methods listed above, the surface shall be cleaned of dirt, grease, and soluble salts prior to treatment. Heavier layers of rust shall be removed by chipping, before any of the other cleaning tools are used.

Finally, all traces of dust shall be removed prior to coating.

216.07.05 Abrasive Blast Cleaning – Abrasive blast cleaning is a procedure for removing mill scale, rust, rust scale, coating or other foreign matter by use of abrasives propelled through nozzles or by centrifugal wheels. It is also used to roughen the surface of the substrate to enhance adhesion.

If compressed air blasting is employed, proper line filters and dryers, which remove all oil and water from the air, shall be used. Compressed air shall be periodically tested for presence of these contaminants.

The standard of cleaning obtained by this method shall be SSPC SP 5, SP 6, SP 7, SP 10, or SP 14 as specified in the Drawings and Special Provisions. Where the standard of cleaning is not defined in the Drawings and Special Provisions, the manufacturer's specifications shall be used.

Prior to abrasive blast cleaning, the surface shall be cleaned of dirt, oil, grease and residual ions in accordance with SS 216.07.02. If necessary, heavier layers of rust shall be removed by chipping before the blast cleaning operation commences.

A cleaned and blasted surface shall be coated within a maximum of 8 hours of cleaning or less depending on humidity, temperature or other factors. The surface immediately prior to coating shall meet the specification in its entirety. The Ministry shall be the sole arbitrator on determining the minimum time between surface preparation and coating. If the surface preparation criteria are not met immediately prior to coating, the surface shall be rejected and re-prepared in accordance with the specification.

Finally, all traces of dust and sand, etc. shall be removed prior to coating.

216.07.06 Abrasive Materials – Abrasive materials, unless specified, must be accepted by the Ministry Representative. Silica Sand shall only be used in accordance with the Workers Compensation Act and the Occupational Health and Safety Regulation.

216.07.07 Recycled Steel Abrasive – Recycled steel abrasives are permitted; however the working material must meet SSPC-AB 2 requirements and must contain a minimum 15% steel grit to produce an angular/sharp profile.

216.07.08 Containment of Debris and Disposal of Debris – All debris from cleaning, blasting and coating operations must be contained in such a way as to not contaminate the environment. All debris shall be contained, tested and disposed of in accordance with current environmental, fisheries and local regulations. The Contractor shall provide written proof of compliant disposal of all materials including waybills and transport logs.

216.08 Application of Coating Materials

216.08.01 Pretreatment – Newly cleaned surfaces may require pretreatment other than washing or abrasive blasting prior to the application of the prime coat of paint. The application of the pretreatment shall be strictly in accordance with the manufacturer's directions. When pretreatments are used, particular care shall be taken to prevent contamination of adjacent surfaces.

216.08.02 General – In general, the Contractor shall follow SSPC-PA 1 Shop, Field, and Maintenance Painting of Steel when applying coatings. Where this SS 216 and SSPC-PA 1 disagree, SS 216 shall be taken as applicable.

216.08.03 Materials Storage and Handling – Materials handling and use shall be as per SSPC-PA 1. Coatings shall be stored in a secure building on-site, but not on the structure itself. Coatings taken onto the structure in cans, pails or totes shall be contained so that should they spill, no coating will escape to contaminate the adjacent area or environment.

216.08.04 Temperature – Coatings shall only be applied when the ambient, surface and coating temperature are within the range indicated by the manufacturer's written instructions. Coating shall not be applied when the substrate temperature is less than 3°C above the Dew point. Winter or quick dry catalysts or coating formulations shall not be used. Some coatings are designed for use in marginal conditions. Their use is at the sole discretion of the Ministry Representative. In unique situations where coatings must be used outside these limits, they shall only be used at the sole discretion of the Ministry Representative. Lack of permission may necessitate removal, re-cleaning and reapplication at the Contractor's expense.

216.08.05 Humidity – Coatings shall not be applied in wet weather, fog, mist, rain, snow and/or relative humidity in excess of 85%. Coatings shall not be applied to wet, damp or icy surfaces and care shall be taken to ensure that any overnight dew has completely evaporated before coating commences in the morning.

216.08.06 Mixing and Thinning – All ingredients in any container of paint shall be thoroughly mixed before use and shall be agitated often enough during application to keep the paint in a uniform condition. In all cases of heavily pigmented paints (e.g. "Zinc Rich" paints) the material shall be continuously agitated during application.

Mixing shall be done in accordance with the manufacturer's instructions and in accordance with good painting practice as outlined in SSPC Steel Structures Painting Manual, Volume 1, Good Painting Practice and SSPC-PA 1. No coating system shall be applied beyond one year from the manufacturing date.

Thinner shall only be added with the written consent of the manufacturer and only after it is shown that satisfactory application cannot be obtained with proper adjustment of the spray equipment and air pressures. In no case shall more than 12.5% of thinner by volume be added to a paint.

Only thinner specified and supplied by the manufacturer for that product shall be added to a product. Thinners used for cleanup shall not be recycled as thinning agents. Paint shall be mixed by mechanical methods. Paint shall be mixed in a manner that will ensure breaking up of all lumps, complete dispersion of settled pigment and a uniform composition. After mixing, all paint shall be strained to remove skins and other undesirable matter; the strainer shall not be so fine as to remove any pigment.

Thinner, when approved by the Ministry Representative, shall be added after initial mixing is completed. The quantity of thinner added shall be measured in and thoroughly mixed until the paint is again of uniform consistency.

216.08.07 Induction Time – Manufacturers' induction times shall be strictly observed before the coating is applied. Since induction times are a function of temperature, temperatures outside the normal range may necessitate changing the induction period. Revised induction times may be accepted by the Ministry Representative, subject to the coating manufacturer providing the revised induction times in writing.

216.08.08 Pot Life – Manufacturer's estimates of pot life shall be adhered to, and no coating shall be used after the recommended pot life has expired. Since pot life is a function of temperature, in hot or cold conditions the pot life shall be revised to reflect the ambient temperature. Written revised pot life recommendations from the coating manufacturer must be reviewed and accepted by the Ministry Representative prior to their use.

216.08.09 Wind – Paint shall not be sprayed when the wind speed or wind gusts are over 15 km per hour. Wind speed shall be measured where the work is taking place, for example within the containment.

216.08.10 Curing Conditions – Coating lifetime may be affected by curing conditions. Coatings shall be protected from rain during their curing. Coatings shall be cured under the coating manufacturer's stated conditions. This may require heating, cooling, humidity or ventilation adjustments to bring the ambient conditions into line with the coating manufacturer's recommended conditions for

curing. In enclosed spaces, ventilation must be provided to remove evaporation vapours from the coating.

Failure to achieve proper curing or curing conditions may necessitate removal, re-cleaning, reapplication or repair of the coating at the Contractor's expense.

216.09 Inspection

216.09.01 General – The Contractor shall provide the quality control functions on the Contract work. Quality assurance will be done by the Ministry. The Contractor shall produce a quality plan for all work to be done and submit this plan for the Ministry's review and acceptance. The Ministry will audit the quality process via the Ministry Representative.

216.09.02 Quality Control Inspection – Quality Control inspectors shall be a third-party qualified NACE CIP Level 2, or SSPC BCI Level 2 (i.e. Certified Coating Inspectors independent from the applicator). Inspection records produced shall be forwarded to the Ministry Representative daily. The records shall be electronic files, preferably Microsoft Excel files, and shall be forwarded to the Ministry Representative in that format to speed dissemination and aid any data manipulation required for the quality audit. Where this is not possible due to lack of communication technology, the records may be hard copy or hand delivered electronic files. The Quality Control inspector(s) shall follow the quality plan as accepted by the Ministry.

All washing and cleaning work shall be inspected by the Contractor before any coating is applied. Each coat of paint shall be inspected by the Contractor and passed by the Quality Control Inspector before the succeeding coat is applied.

Coated surfaces rejected shall be made to comply with all requirements of this specification at the sole expense of the Contractor.

(a) Hold Points – Hold points (as defined in SS 145.12) for inspection shall include, but may not be limited to:

- **(i)** Post installation of scaffolding and other accesses
- **(ii)** Post installation of containment
- **(iii)** On site coating systems review (manufacturing dates/batch numbers, etc.)
- **(iv)** Surface after washing
- **(v)** Surface after cleaning prior to coating application
- **(vi)** Coating product mixing/thinning
- **(vii)** Surface after Prime coat
- **(viii)** Surface after Penetrating Sealer if any
- **(ix)** Surface after Stripe Coat
- **(x)** Surface after each midcoat
- **(xi)** Surface after caulking if any
- **(xii)** Surface after topcoat
- **(xiii)** Post removal of containment
- **(xiv)** Post removal of scaffolding and other accesses and repair of any coating damage

(b) Accessibility for Inspection – The Contractor shall provide the Ministry Representative safe access to the specific areas of the jobsite under inspection. In order to limit the work disruption, the Contractor shall notify the Ministry Representative of upcoming hold points at least 24 hours prior to the hold point. The Ministry may perform quality assurance/audit inspections in conjunction with or subsequent to the Contractor's quality control inspections. The Ministry Representative will attempt to limit delays to the Contractor as much as possible, but retains the right to obtain as much information from the quality assurance/audit inspections as needed to justify acceptance or rejection of the Work.

The Contractor shall maintain a minimum 2150 lumen/m^2 (200 foot-candles) illumination at all areas of the work to be inspected. This may include adding lighting to augment natural light. Lighting if added shall be according to SSPC Guide 12, Guide for Illumination of Industrial Painting Projects.

216.09.03 Thickness Measurement – Large areas amenable to the application of SSPC-PA 2 shall be measured by this method. The use of PA 2 is not considered to limit the number of measurements should additional readings be needed for any reason.

The Contractor shall set aside a suitable area of steel on the structure to be coated and dry abrasive blast clean the area with the Contractor's equipment and blast abrasive to the standard required by the coating specification. This area shall be protected from moisture and shall be used as a standard for acceptable surface cleanliness and anchor profile. A portion of the area shall also be used as an area to verify the calibration of dry film thickness (DFT) gauges. It is recognized that the magnetic plane on the blast cleaned surface is above the bottom of the profile. The Contractor shall fill that difference with coating. If requested by the Ministry Representative, the Contractor shall produce and calibrate a secondary, portable calibration standard for dry film thickness gauges. The dry film thickness of small and/or more complex areas shall be measured as agreed upon by the Contractor's Inspector and the Ministry Representative. Gauges are to be calibrated according to SSPC-PA 2 at the beginning and end of each work shift.

216.09.04 Tinting Multilayer Systems – Each separate coat, penetrant, stripe, midcoat(s) and topcoat shall be coloured or tinted to differentiate it visually from the coats above and below. This is an aid to application and

inspection and also serves as an indicator of wear for the life of the coating.

216.09.05 Coating Systems – SS 308 lists coating systems used or identified by the Ministry and suggests situations where the systems are best applied. Coating systems meeting SS 308 requirements are listed in the Recognized Products List. Only specified coating systems from the Recognized Products List shall be used.

Occasionally, special requirements mandate the use of coating materials not already listed in SS 308. These requirements will normally be specified in the Contract Special Provisions section or will be shown on the Contract Drawings.

Unless otherwise specified, surface preparation, application and curing for individual coatings shall be in accordance with the manufacturer's Product Data Sheet. Coating thickness shall be within the range recommended by the manufacturer's Product Data Sheet as judged by SSPC PA-2 Level 3 inspection unless otherwise specified.

216.09.06 Coatings from Different Manufacturers – The Contractor shall not mix or interchange coatings or solvents from different manufacturers.

216.09.07 Coating Flaws – A coat of paint will only be accepted if none of the following faults are apparent:

(a) Coatings showing sags, runs, holidays, pinholes, orange peel, dry spray, shadowing or other application defects.

(b) Poor coverage at rivet heads, bolts, threads, plate edges, crevices, pockets, corners or re-entrant angles.

(c) Surfaces damaged mechanically by moving scaffolding, equipment, etc. or damage by rain, flying dust, etc. or damage from any other cause.

Sags and runs may be repaired or treated as per SSPC-PA 1.

Coatings shall be applied as per SSPC-PA 1, except where directed otherwise by this specification.

216.09.08 Rejection – Areas not meeting this specification shall be repaired at the sole expense of the Contractor. At the sole discretion of the Ministry Representative, small areas may be spot repaired. Larger areas shall be re-cleaned to specification and the coating built up again to specification.

216.10 Application Methods (Excluding metallizing and Hot Dip Galvanizing)

216.10.01 General – Coatings are designed to be applied by specific application techniques. These techniques are specified by the manufacturer to ensure that the risk of failure is minimized. Manufacturer's product application sheets shall be followed when applying coatings.

Where an application method is not possible due to circumstances at a particular structure, the Contractor shall obtain written instructions from the manufacturer for a new application method. Contractor use of the new application method is subject to approval by the Ministry Representative.

216.10.02 Brush or Roller Application – Brushes or rollers shall be of high quality and of a style and material that will enable proper coating application. The paint shall be brushed or rolled so that there are no runs or sags, minimum brush marks and a uniform thickness. Achieving the specified thickness for a coating may require more than one coat with brush or roller. The Contractor is responsible for achieving the specified thickness.

216.10.03 Air Spray Application – The equipment shall be capable of properly atomizing the paint to be applied. The equipment shall be kept clean and maintained in excellent condition. Equipment that has dirty or malfunctioning parts shall not be used until the parts are cleaned, repaired or replaced as necessary.

Paint shall be applied in a uniform layer, with overlapping at the edge of the spray pattern. The spray pattern shall be adjusted so that the paint is deposited uniformly. During application, the gun shall be held perpendicular to the surface and at a distance that will ensure that a wet layer of paint is deposited on the surface. The trigger of the gun shall be released at the end of each stroke.

All runs and sags shall be brushed out immediately, or the paint shall be removed and the surface repainted.

Areas inaccessible by spray such as cracks and crevices and the blind sides of rivets and bolts shall be painted by brush. Stripe coat material may be applied by spray to speed up the transfer of material to the surface, but all stripe coating must be brushed in to ensure complete wetting and coverage.

The air caps, nozzles and needles shall be those recommended by the manufacturers of both the material being sprayed and the equipment being used.

Proper air line filters and dryers shall be provided to remove oil and condensed water from the air. The air from the spray gun impinging against the surface and into a clean rag or blotter paper shall show no condensed water or oils.

The pressure on the material in the pot and of the atomizing air shall be adjusted for optimum spraying effectiveness. Both material and atomization pressures shall be controlled by suitable pressure regulators with gauges. The atomizing air pressure shall be high enough to properly atomize the paint, but not so high as to cause excessive fogging of paint, excessive evaporation of solvent or loss by overspray.

The air compressor used shall be of sufficient size to maintain the regulated air pressures constant at all times, irrespective of the amount of equipment that is being operated from the compressor.

216.10.04 Airless Spray Application – The equipment shall be capable of properly atomizing the paint to be

applied. The equipment shall be kept clean and maintained in good working condition. Equipment that has dirty or malfunctioning parts shall not be used until the parts are cleaned, repaired or replaced as necessary.

Paint shall be applied in a uniform layer, with overlapping at the edge of the spray pattern. The spray pattern shall be adjusted so that the paint is deposited uniformly. During application, the gun shall be held perpendicular to the surface and at a distance that will ensure that a wet layer of paint is deposited on the surface. The trigger of the gun shall be released at the end of each stroke.

All runs and sags shall be brushed out immediately, or the paint shall be removed and the surface repainted.

Areas inaccessible by spray such as cracks and crevices and the blind sides of rivets and bolts shall be painted by brush. Stripe coat material may be applied by spray, to speed up the transfer of material to the surface, but all stripe coating must be brushed in to ensure complete coverage.

Fluid tips shall be of proper orifice size and fan angle, and fluid control gun of proper construction as recommended by the manufacturers of both the material being sprayed and the equipment being used.

The air pressure to the paint pump shall be adjusted so that the paint pressure to the gun is proper for optimum spraying effectiveness. The pump pressure shall be controlled by a suitable regulator with working gauges.

Pressures higher than that necessary to properly atomize the paint shall not be used.

The air compressor used shall be of sufficient size to maintain the regulated air pressure to the pump constant at all times irrespective of the amount of equipment being operated from the compressor.

Proper air line filters and dryers shall be provided to remove oil and condensed water from the air. The air from the spray gun impinging against the surface and into a clean rag or blotter paper shall show no condensed water or oils.

The trigger of the gun shall be pulled fully open and held fully open during all spraying to ensure proper application of paint. The trigger shall be released at the end of each stroke.

216.10.05 Mitt and Dauber Application – Mitt and dauber application shall be limited to situations where other coating application methods are poorly suited. Their use shall be at the sole discretion of the Ministry Representative.

Coating thickness shall be uniform over the entire surface of the object coated. Coating shall be free of pinholes or other holidays, sags, etc.

216.10.06 Powder Coatings – Powder coatings are finely ground coatings that are electrostatically sprayed onto the substrate, then melted into a consolidated layer by heat. Coatings may be thermoplastic or thermosetting. Cleaning is typically in a series of chemical baths.

Substrates contaminated with soluble salts shall be washed to remove those salts before being cleaned to meet specification. Acceptable levels of soluble salts shall be at the "designated NVC levels" as stated in SS 216.07.02(b).

The substrate to be coated shall be cleaned to the coating manufacturer's specifications. The water test shall show a clean surface.

If required for adhesion, a conversion coating shall be applied to the substrate before powder coating.

Fresh and reclaimed powder shall be dry, free flowing, and free of lumps and contaminants.

Soaking oven temperatures and residence times shall be sufficient to completely melt the powder, but not enough to oxidize or discolour the coating.

Coating thickness and gloss shall be uniform over the entire surface of the object being coated. Coating shall be free of pinholes or other holidays.

216.11 Application Methods (Metallizing and Hot Dip Galvanizing)

216.11.01 Metallizing – Metallizing shall be applied in accordance with SSPC-CS 23.00/AWS C2.23M/NACE No. 12.

Metallizing includes the thermal spraying of zinc, aluminum or their alloys onto a properly prepared metal substrate.

The method of heating the spray metal is either by flame or by plasma arc. Spray metal may be in wire or powder form to fit the apparatus used to spray the metal. The application method shall ensure that a uniform, clean, well adhered, film of metal is applied to the steel surface.

Surface preparation shall be SSPC SP5/NACE No. 1. Profile shall be sharp, 50 to 75 µm. Cut metal edges may be harder and consequently have less profile. If that is the case, they shall be ground back to softer metal and reblasted to achieve the specified profile.

Metallizing shall be applied by qualified personnel with documented field experience metallizing steel structures such as bridges.

Metallizing shall be applied in a uniform layer at a thickness of 150 to 200 µm thickness. For special applications, the thickness may be specified outside this range.

When the metallizing is to be overcoated to form a duplex system, the newly metallized surface shall be protected from contamination until coated. If the metallized surface becomes contaminated, the steel substrate shall be re-cleaned to SSPC SP5/NACE No. 1 as above and re-metallized.

Metallizing is porous, and application of a coating over the surface may require a mist coat or other special procedures to avoid bubbling and subsequent problems caused by the interaction of the porous coat with the coating being applied. This cost, if any, shall be borne by the Contractor.

Metallizing may be applied over a much wider range of ambient conditions than normal paint. When application conditions deviate outside normal coating conditions, the Contractor shall consult the Ministry Representative for acceptance before proceeding.

216.11.02 Hot Dip Galvanizing – Hot dip galvanizing shall be applied in accordance with ASTM A123 or ASTM A153, as applicable.

216.12 Application of Coatings

216.12.01 General – Application of coatings on steelwork specified to be painted as part of a Contract shall meet the following requirements.

(a) Welding Requirements – Areas to be field welded shall have no paint applied for a distance of at least 100 mm back from the weld area.

If inorganic zinc primer is used, the set-back shall be a minimum 25 mm.

All welds shall be cleaned of weld residue including weld spatter and roughness on the welds. All porosity and pin holes in welds shall be repaired by welding. Sharp edges shall be rounded to a minimum radius of 1 mm. Burrs around drilled holes shall be removed.

All welds shall receive a stripe coat of midcoat material over the primer prior to the midcoat application.

Welding done on existing coated surfaces shall have the coating removed, past the heat affected zone or 100 mm from the weld whichever is more, down to bare metal. Surface profile of the cleaned area shall be 50-75 µm (2-3 mils).

(b) Faying Surfaces – Slip-critical bolted connection locations shall be as identified on the Drawings, Special Provisions or as directed by the Ministry Representative.

In the parts of a structure where steelwork is specified to be painted, the following shall apply:

(i) General – Over thickness will not be accepted. If over thickness occurs, then the surface will be cleaned and recoated at the Contractor's expense.

Touch up coating and partial repairs of faying surfaces shall not be permitted.

(ii) For Slip-Critical Connections

(A) Faying surfaces shall be blast cleaned and shall receive one coat of inorganic zinc primer. No midcoat or topcoat shall be applied.

(B) Primer coatings shall meet the Class B coating requirements as specified in CSA-S6.

(C) The class of the coating shall be determined based on the results from testing done in accordance with the "Specifications for Structural Joints Using High-Strength Bolts" issued by the Research Council on Structural Connections.

(D) A copy of the certificate of testing, the test procedures and results, and the test specimen preparation procedures for the Class B coating shall be supplied to the Ministry Representative for each coating used on faying surfaces.

(E) Faying surfaces shall have the same blast cleaning and coating application as was used in the tests to determine the mean slip coefficient. Coated joints shall not be assembled before the coating has cured for the minimum time used in the tests to determine the mean slip coefficient.

(iii) For Connections that are Not Slip-Critical

(A) Faying surfaces shall be blast cleaned to SSPC-SP 5/NACE No. 1 and receive one coat of inorganic zinc primer. No mid or top coat shall be applied.

(B) Primers shall be applied and cured in accordance with this SS 216.

(c) Surfaces Inaccessible After Erection – Surfaces inaccessible after erection shall receive the field coating in the shop, before shipment to the erection site. Prior to erection, handling damage shall be repaired to the satisfaction of the Ministry Representative.

(d) Application and Curing of Successive Layers of Coating – Each layer of coating shall be in a proper state of cure or dryness before the application of the succeeding coat. Unless otherwise specified a minimum of overnight drying shall be required. Under poor drying and curing conditions, this period will be extended.

Curing conditions shall be those given by the coating manufacturer in the product application instructions. Curing time will vary according to temperature and in some cases humidity. The required curing time is estimated at the ambient conditions during curing. This will normally be done from data supplied by the coating manufacturer.

Coatings that have a maximum overcoating window shall not be overcoated after that time without

appropriate surface preparation as provided in writing by the coating manufacturer. The overcoat window will be calculated from data supplied by the coating manufacturer using the ambient conditions at the Site.

(e) Primer – Primer shall not be applied until the cleaned steel surface meets specification requirements. Once specification requirements are met for a substrate, the primer must be applied within 8 hours. The substrate if protected will be assumed to continue to meet the specification for 8 hours unless obvious contamination has occurred. Determination of this will be at the sole discretion of the Ministry Representative.

Primer shall be applied to specification requirements. Should the specification requirements be different than the coating manufacturer's application instructions then the more stringent specification shall apply unless otherwise approved by the Ministry Representative.

(f) Penetrating Sealers – Penetrating sealers or low viscosity products designed to wick into seams and seal them shall be applied according to the coating manufacturer's product data sheets unless the Contract calls for specific application parameters. Should that be the case, the sealer shall be applied to the Contract requirements.

Current industry thinking has these products applied on top of a single primer coat and under stripe and midcoats.

Under certain circumstances it may be necessary to use a penetrating sealer as a primer or as a first coat. This will be at the sole discretion of the Ministry Representative.

(g) Stripe Coat – All areas such as crevices, corners, rivets, bolts, nuts, welds, edges and other protrusions shall be given a stripe coat to ensure that there is sufficient coating dry film thickness. Unless it is specifically excluded, it is the intent of the Ministry that all such areas shall receive a stripe coat. Coating may be applied by spray to save time, but it shall be brushed into all areas to be striped.

The stripe coat is a separate coat and shall be applied at the thicknesses given for the midcoat coating.

The stripe coat shall be applied to the area and at least 25 mm out from the area to be striped.

The stripe coat shall be a contrasting colour to the coats above and below it.

The stripe coat shall be cured before being overcoated.

(h) Midcoat – Midcoat shall be applied as a full coat onto clean, cured, primer, sealer, and stripe coats. The stripe coat and or sealer shall be cured before being coated with a subsequent coat.

(i) Caulking – Caulking shall be applied after the midcoat and prior to the topcoat.

Caulking shall be compatible with the coating system used.

Silicone Caulking shall not be used.

Suitable caulking materials shall be in accordance with SS 308 and as listed in the RPL under the sub heading System PCM Paint Caulking Materials.

Caulking shall be applied as specified in the Contract.

Caulking shall not be top coated until it cures according to the manufacturer's product data sheets. Lower temperatures may increase this time.

If there is a time window for top coating, the caulking shall be top coated before the window expires.

(j) Topcoat – Once the undercoats and caulking if any have cured, the topcoat shall be applied.

(k) Lapping Existing Coatings – Primer if applied shall not overlap existing coatings by more than 10-20 mm. Ministry preference is that it overlaps very slightly to seal the interface joint.

Midcoat shall overlap existing coatings by a minimum 50 mm.

Topcoat shall overlap existing coatings by a minimum 50 mm and shall hide all newly added midcoat material.

216.12.02 Shop Coating – Unless otherwise specified, the following procedure shall apply to all new steel prior to shipment to the site of erection:

Before blast cleaning, the steel shall be checked for contamination and if necessary the surface cleaned in accordance with SS 216.07.02.

Before coating, the steel shall be blast cleaned to SSPC-SP 10/NACE No. 2 Near white blast. Abrasive blast cleaning will be permitted prior to fabrication provided that any contamination incurred during fabrication is removed prior to coating, leaving an SSPC-SP 10/NACE No. 2 surface. Small areas of contamination occurring during fabrication shall be cleaned by manual cleaning to SSPC-SP 3 Power tool cleaning and/or SSPC SP WJ-4/NACE WJ-4 as necessary to remove contamination. Large areas of contamination occurring during fabrication shall be dry abrasive blast cleaned again to SSPC-SP 10/NACE No. 2 Near-white blast.

Surface profile for primers shall be 50 to 75 µm and sharp.

Prior to dry abrasive blasting, all welding slag, spatter and alkaline deposits in the vicinity of the welds shall be removed. Surfaces shall then be blast cleaned to SSPC-SP 10/NACE No. 2 Near-white blast.

When fabrication is completed and the surfaces cleaned as required, they shall be inspected for radius of corners, weld spatter, unwelded interfaces, porosity or pin holes in welding and burrs around drilled holes prior to the application of paint.

After passing inspection and before any contamination has occurred, the Contactor shall prime all steel surfaces with one coat of the selected shop coat primer from SS 308. Primer shall be applied to specification requirements. A dry abrasive blast cleaned surface shall be coated within 8 hours of cleaning and before rust blooming becomes apparent. If either condition is exceeded then the area must be re-blasted to specification.

Clean or partially coated steel shall be protected from contamination including, but not limited to dirt, oil, water, grease, soluble salts and overspray, while in the shop. Contaminated surfaces shall be re-cleaned at the Contractor's expense.

Surfaces that will be in contact with concrete shall be coated with the full system prior to erection. Coatings with known incompatibilities with high pH shall not be used in contact with the concrete.

Surfaces inaccessible after erection shall receive the field coats of paint in the shop, before shipment to the erection site.

Faying surfaces shall meet the requirements of SS 216.12.01(b).

If more than one coat is to be shop applied, the manufacturer's recommended recoat times for the ambient temperatures shall be followed.

Generally, a minimum of 16 hours drying/curing time shall be required between the application of any paint in the shop and the moving or handling of any piece so painted. This period may be increased in cases where poor drying conditions exist, such as shop areas open to the weather in winter months. Since curing times vary with different coating products, variations from this time may occur with specific products. Blocking used to support structures during the coating applications will be moved after the curing of applied paint is complete and the surface area of the original blocking locations shall be prepared and painted as required by this specification.

Quick Dry/Cure products or products with "winter" catalysts shall not be substituted for SS 308 products unless accepted by the Ministry Representative.

216.12.03 Transport of Coated Items – Transport of coated items shall be done in a manner that imparts minimum damage to the coating system.

The coatings shall be cured before handling and shipment.

Canvas or Fabric slings shall be used to move the items. Chains or other methods that are known to damage coatings shall not be used.

Coated items to be transported shall have sufficient dunnage to prevent coating damage. Transport tie-downs shall be padded to protect the coated surfaces and edges.

Transport damage shall be repaired at the Contractor's expense.

216.12.04 Field Repair of Damaged Shop Coating and Field Weld Areas – After the complete erection of shop coated steel members the following procedure shall apply:

Before coating, field weld areas shall be ground off to remove all welding slag, spatter and alkaline deposits in the vicinity of the welds. Washing will be required when the shop coat or steel has been contaminated with air borne dirt, dust, salt, chemicals, etc., unless the steel shows no visible contamination, and meets the "designated NVC levels" as stated in SS 216.07.02(b) for non-visible contaminants. Surfaces shall then be cleaned to SSPC-SP 11 . The existing coating shall be feathered back 25 mm from damaged areas into good coating. Feathering shall continue until the existing coating is well adhered. This may be considerably more than the 25 mm feather back.

Care shall be taken not to damage surrounding areas.

Note: This cleaning shall include the 100 mm welding set back.

All field connections and all areas of the shop applied coating system damaged during handling, shipping and erection shall be cleaned to SSPC-SP 3 or SSPC SP 11 at the discretion of the Ministry Representative. Low pressure water cleaning shall be used as necessary to remove dirt, salts and other deleterious contamination.

All areas that have been cleaned to bare metal as required shall be coated with an organic zinc field primer, applied by brush to ensure full coverage. If the area is too large to coat practically with a brush, then spray may be used, providing overspray does not contaminate the surrounding finished coating.

Lapping of new and existing coatings shall be as specified in SS 216.12.01(k).

If a partial system was applied in the shop, then the primed areas shall be coated with successive coats to the level of the surrounding coating and then the entire structure shall be top coated. This treatment shall include a brushed-in stripe coat.

If a coating system was completely applied in the shop and on erection some damage has occurred or there are still uncoated areas, then these areas shall be cleaned in accordance with this section and successively coated with each coat of the paint system up to the topcoat including a brushed-in stripe coat. The manufacturer's recoat times for the ambient temperatures shall be followed.

216.12.05 Field Coating – Remaining coats shall be applied as specified.

Faying surfaces shall meet the requirements of SS 216.12.01(b).

(a) Touch-up coating- This section deals with the repair of small areas of coating. The definition of small areas will vary depending on the situation but, in general, will apply to areas that can be economically treated using the touch-up technique.

The coating used for touch up and repair shall be one of the SS 308 Field Overcoat (FO) systems unless otherwise specified.

The area to be repaired and coated shall be washed in accordance with SSPC SP WJ-4/NACE WJ-4, to achieve a surface cleanliness as per SS 216.07.02(b) before attempting further cleaning. If the area meets the "designated NVC levels" as per SS 216.07.02(b) and is free of dirt and detritus, washing may be omitted. Areas of the coating that have failed or rusted shall be cleaned to SSPC-SP 3 or SSPC-SP 11 using the appropriate tools.

The old coating shall be feathered back 25 mm from damaged areas into good coating. Feathering back shall continue until the coating remaining is well adhered. This may be considerably more than the 25 mm feather back.

Pack rusted crevices will require more thorough cleaning. The degree of cleaning required will be detailed in the Contract Documents for each specific structure. After cleaning, they shall be coated with the prime coat and then a brush coat of penetrating sealer shall be applied to the crevice and allowed to soak in. Repeated applications of the penetrating primer may be necessary until the inside of the crevice is saturated. Where the coating system used does not include a specific primer coat, the penetrating sealer shall be applied as a first coat.

In general, coating used for touch up and repair shall be of the same type as those used initially. For example if the coating system to be touched up is an alkyd system, then alkyds are the coatings of choice. The coating to be used shall be subject to acceptance by the Ministry Representative or as specified in the Contract.

The selected primer shall be applied in the damaged areas. The primer shall be cured for the time recommended by the manufacturer prior to additional coating being applied

After the primer is cured, the midcoat(s) shall be applied to bring the surface of the coating to the same level as the surrounding sound coating. The midcoat shall be applied within the manufacturer's suggested range of thickness. The midcoat(s) shall be cured for the time(s) recommended by the manufacturer.

After the midcoat material is cured, the area shall be given a topcoat. Where practical, the topcoat shall be extended to a logical edge such as the edge of the beam or plate, to present a less patchy appearance.

Lapping of new and existing coatings shall be done in accordance with SS 216.12.01(k).

(b) Overcoating – The object is to clean the surface, repair spot or area damage with coatings and to overcoat the entire area with a topcoat.

The coating used for touch up and repair shall be one of the SS 308 Field Overcoat (FO) systems.

The area to be repaired and coated shall be washed to the requirements of SSPC-SP WJ-4/NACE WJ-4, with a surface cleanliness meeting the "designated NVC levels" as per SS 216.07.02(b) before attempting further cleaning. If the area meets the "designated NVC levels" as per SS 216.07.02(b) and is free of dirt and detritus, washing may be omitted. Areas of the coating that have failed or rusted shall be cleaned to SSPC-SP 3 or SSPC-SP 11. Larger failed areas shall be cleaned to SSPC-SP 6/NACE No. 3. In this case, surface profile should be 50-75 µm and sharp.

Debris contaminating the rest of the structure or the immediate environment due to cleaning operations shall be removed by the Contractor.

The old coating shall be feathered back 25 mm from damaged areas into tightly adhered coating. If the coating cannot be removed with a dull putty knife, it is considered tightly adhered.

Pack rusted crevices shall be cleaned in accordance with SS 216.12.05(a).

Once all the existing damaged areas are built up to the level of the existing coating, the topcoat is applied over the entire area to seal the existing coating and all the repairs.

In general, coating used for overcoating shall be of the same type as those used initially. For example if the coating system to be touched up is an alkyd system, then alkyds are the coatings of choice. Use of an over wintered test patch is recommended to test compatibility. The coating to be used shall be subject to acceptance by the Ministry Representative or as specified in the Contract.

(c) Recoating – The area to be recoated shall be washed to the requirements of SSPC-SP WJ-4/NACE WJ-4, to achieve a surface cleanliness meeting the "designated NVC levels" as per SS 216.07.02(b) before attempting further cleaning. The entire area shall be blast cleaned to SSPC-SP 6/NACE No. 3. Note that some specialty coatings or coatings in some

difficult applications such as immersion may require SSPC-SP 10/NACE No. 2 or even SSPC-SP 5/NACE No. 1.

Surface profile should be 50 – 75 µm and sharp.

The Contractor shall provide full containment of debris generated by the cleaning, coating and curing operations.

Debris contaminating the rest of the structure or the immediate environment due to cleaning operations shall be removed by the Contractor.

Sharp edges shall be radiused to 2 mm.

(d) Zone Coating – Zone coating may use any one of or a combination of touch-up, overcoating or recoating on limited areas of a structure.

(e) Field Curing of Coatings – Each coating application shall be cured before it is coated over or exposed to weather or traffic.

Coatings inside enclosed spaces shall continue to be ventilated until dry or cured to ensure that they cure correctly.

Uncured coating shall be protected against damage from dust or other detrimental foreign matter. If necessary, coating shall be delayed until the cause of the dust, etc. has abated. Uncured coating shall be protected from the weather until cured. In some cases, this may include hoarding, heating, ventilation or dehumidification. Direct propane heaters shall not be used for heating during the painting or curing process. Coating contaminated or damaged by any of the above conditions shall be repaired at the Contractor's expense.

216.13 Damage to Adjacent Property – It is the intention of the Ministry that the coating work shall have no adverse effect on adjacent property. Repair of damage to adjacent property, vehicles, pedestrians and other portions of the structure due to the Contractor's coating operations is the sole responsibility of the Contractor.

216.14 Cleanup – All construction and painting debris must be removed from the site and the site environs. The site and surroundings shall be left in the same condition as prior to work being commenced.

216.15 Measurement and Payment – Measurement and Payment shall be as set out in the Contract Documents.

SECTION 218
CONCRETE FOR MINOR WORKS

SCOPE

218.01 General – SS 218 shall be used only where explicitly specified by the Ministry in the Contract.

Concrete produced under SS 218 is intended to be used only for non-structural reinforced and non-reinforced cast-in-place concrete works for Ministry infrastructure such as fence post, guardrail embedment, thrust blocks, and small culvert end walls.

(a) **Class A:** for concrete exposed to chlorides and freezing and thawing.

(b) **Class B:** for continuously submerged concrete exposed to chlorides, but not to freezing and thawing.

(c) **Class C:** for concrete exposed to chlorides, but not to freezing and thawing.

For infrastructure that will be owned by a third-party, such as a municipality, the specifications of that owner will normally apply (such as MMCD).

MATERIALS

218.10 Cement – Cement shall conform to the requirements of CSA A3000 and CSA A23.1.

218.11 Aggregate – Concrete aggregate shall be tested for sulphate content in accordance with CSA A23.2-3B.

218.12 Concrete – Concrete for minor works shall be in accordance with CSA A23.1 – Concrete materials and methods of concrete construction.

Unless otherwise specified, concrete shall meet the requirements of Table 218-A, Class A concrete, and the requirements of the sulphate exposure category identified in the contract.

218.13 Mix Design and Performance – The Contractor shall be responsible for the performance of the concrete, carrying out all testing to verify performance, quality control of all aspects of the concrete operations and for proportioning all concrete mix designs.

The mix shall be designed to be compatible with the concrete aggregate, as well as to meet the sulphate exposure identified in the Contract.

The Contractor shall submit documentation demonstrating that the concrete mix design and constituent components meet the specified requirements and providing the details of the quality control work plan. The documentation shall be submitted to the Ministry Representative at least two (2) weeks in advance of the when concrete production is scheduled to commence.

No concrete shall be placed prior to the completion of the Ministry review and acceptance of the documentation.

PAYMENT

218.90 Basis of Payment – Payment for Concrete for Minor Works is usually incidental to another pay Item.

Accordingly, no direct payment shall be made for Concrete for Minor Works unless the "Schedule of Approximate Quantities and Unit Prices" includes a specific pay Item for Concrete for Minor Works.

When there is a specific pay Item, payment shall be made at the price bid per cubic metre of concrete, measured in-place in and forming part of the finished Work. Payment shall be accepted as full compensation for everything furnished and done in connection therewith.

Table 218-A: Classification of Concrete Mixtures

Class of Concrete	Required Strength at 28 Days (MPa)	CSA A23.1 Class of Exposure*
A	32	C-2
B	30	C-3
C	25	C-4

*Class of exposure shall also be compatible with the sulphate exposure category identified. If no exposure category is identified the class of exposure shall also meet the requirements of S-3.

SECTION 220
SHOTCRETE AND SOIL ANCHOR WALLS

SECTION UNDER DEVELOPMENT

SECTION 232

METAL BIN-WALL

232.01 Scope – This Section covers the construction of metal bin type retaining walls (Metal Bin-Wall) in accordance with the general layout and details indicated on SS Drawings 1-SP232 through 3-SP232.

232.02 Provision of Metal Bin-Wall – Metal Bin-Walls of the design (depth to height ratio) called for shall be constructed at the locations and as shown on the Drawings with the materials, accessories, and necessary ancillary work all in accordance with the details indicated on the Drawings, Specifications, Special Provisions, the SS Drawings or as directed by the Ministry Representative. Any additional installation requirements specified by the Metal Bin-Wall supplier shall also be adhered to by the Contractor.

The Contractor shall provide the manufacturer's drawings (shop drawings) to the Ministry Representative for review at least two (2) weeks prior to components being delivered to Site. The shop drawings shall clearly show the following:

(a) height and depth of all bins;

(b) thickness of all stringers, spacers, verticals and other components;

(c) material specifications;

(d) assembly instructions; and

(e) any special details.

232.03 Materials – Metal Bin-Wall materials will be supplied by the Contractor in accordance with the Contract.

Galvanized steel sheet and galvanized fasteners shall be in accordance with CSA G401. Vertical connectors and base plates shall be in accordance with CSA G40.21 Grade 260W minimum and shall be galvanized in accordance with ASTM A123.

232.04 Construction

232.04.01 Construction – Construction shall be carried out with all labour, tools, equipment and incidentals necessary to complete all Metal Bin-Wall construction in accordance with good work practice.

232.04.02 Foundation – Foundation excavation shall be carried down to the depths shown on the Drawings or to such greater depths as the Ministry Representative may direct. In the case where rock, hardpan or other unyielding material is encountered, it shall be removed to a depth 300 mm below the design grade of the bin-wall footing and backfilled in accordance with SS 232.04.04.

232.04.03 Erection – The units shall be assembled as shown on the drawings. Components shall be handled carefully and any which are damaged shall be removed and replaced with new components at the Contractor's expense.

Minor damage, in the opinion of the Ministry Representative, to the galvanized coating shall be repaired by coating with two heavy coats of zinc rich paint selected from the Ministry's Recognized Products List under the category of "Additional Paint Coatings – Zinc-Rich Touch-up Paints and Primers".

Before the assembly bolts are tightened, columns shall be checked for line and batter or plumbness, where applicable.

232.04.04 Backfill – Structure backfill material shall be in accordance with the material requirements for Bridge End Fill (BEF) as specified in SS 202.

Structure backfill material shall be placed as wall foundation, backfill inside the bins, and for 0.6 m outside the wall. The material shall be spread in layers not exceeding 150 mm in thickness and shall be compacted to a minimum 95% Standard Proctor Maximum Dry Density obtained by the current ASTM test method D698.

Backfill inside each bin shall progress simultaneously but not in advance of that outside the wall.

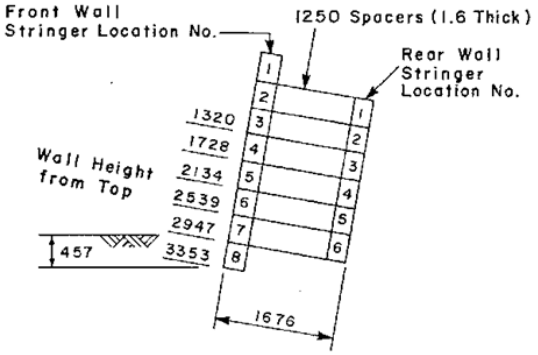
Note: Bin-wall may be backfilled as it is erected provided that the columns are held in correct position while the backfill is being placed.

232.05 Measurement and Payment – Metal Bin-Wall shall be paid for at the Unit Price bid per square metre of facial area of wall, measured on the batter and including the toe burial. Payment shall be full compensation for work required to supply and install the Metal Bin-Wall, including but not limited to, supply, hauling to the Site, and erection of all steel components and also supply, hauling, placement and compaction of the structure backfill.

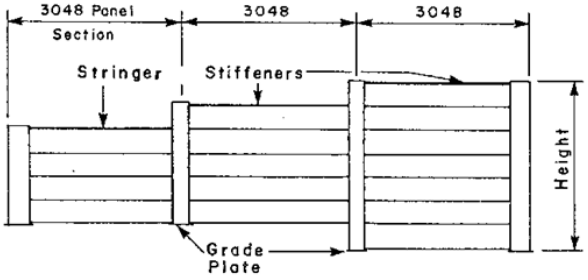
Excavation for foundation shall be measured and paid for in accordance with and at the rate bid for either "Foundation Excavation" in accordance with SS 407 or for "Roadway Drainage and Excavation" in accordance with SS 201, as specified in the Contract Documents.

Backfill placed beyond 0.6 m from the face of the Metal Bin-Wall shall be incidental to "Foundation Excavation" in accordance with SS 407 or "Roadway Drainage and Excavation" in accordance with SS 201.

DESIGN A

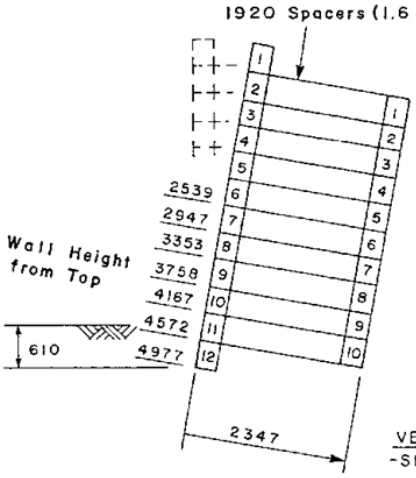


ELEVATION

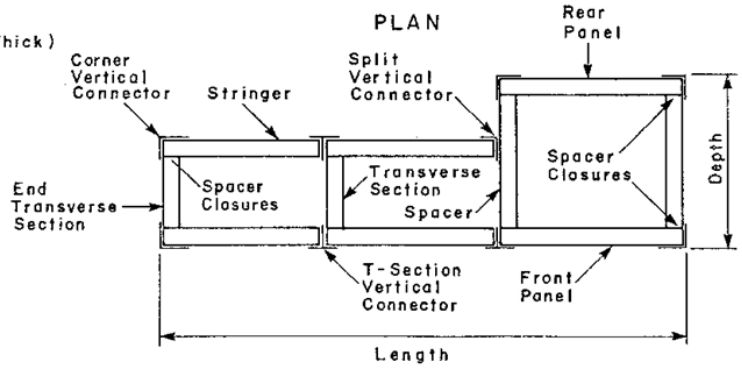


Front Wall Stringer Location No.	Rear Wall Stringer Location No.	Thickness
1 through 8	1 through 6	1.6
9 through 12	7 through 10	2.0
13 through 16	11 through 14	2.8

DESIGN B

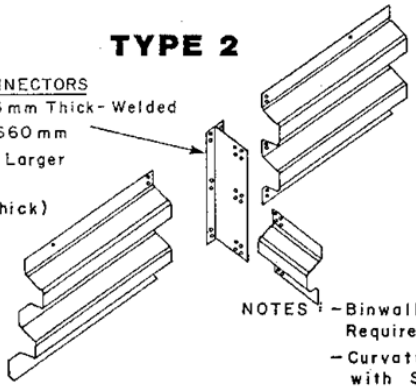


PLAN

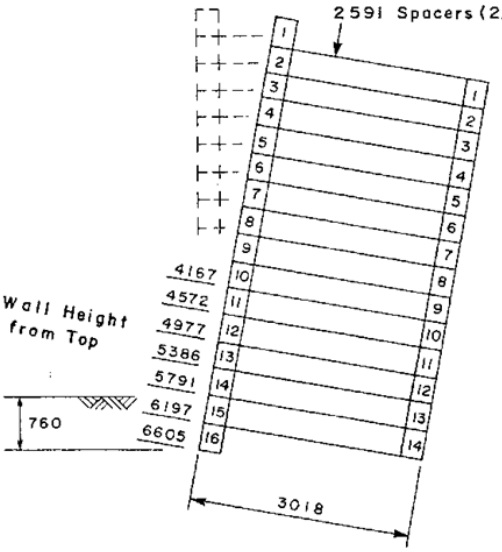


TYPE 2

VERTICAL CONNECTORS
- Steel Plate 6.3 mm Thick - Welded
- Max. Length 3660 mm
- Splice Plate If Larger



DESIGN C

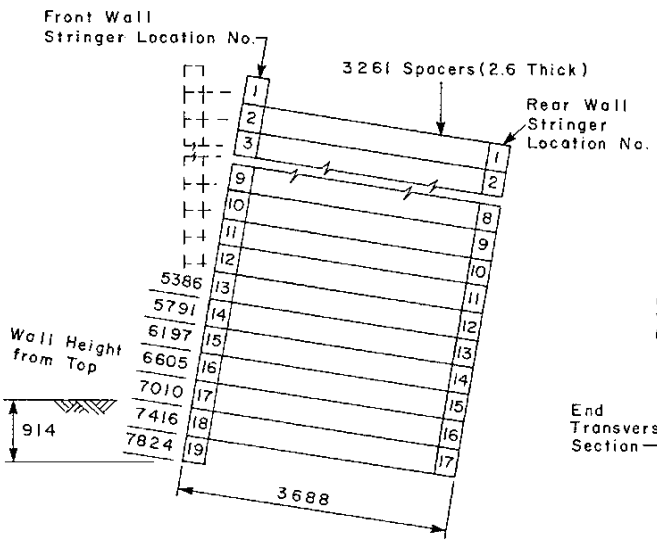


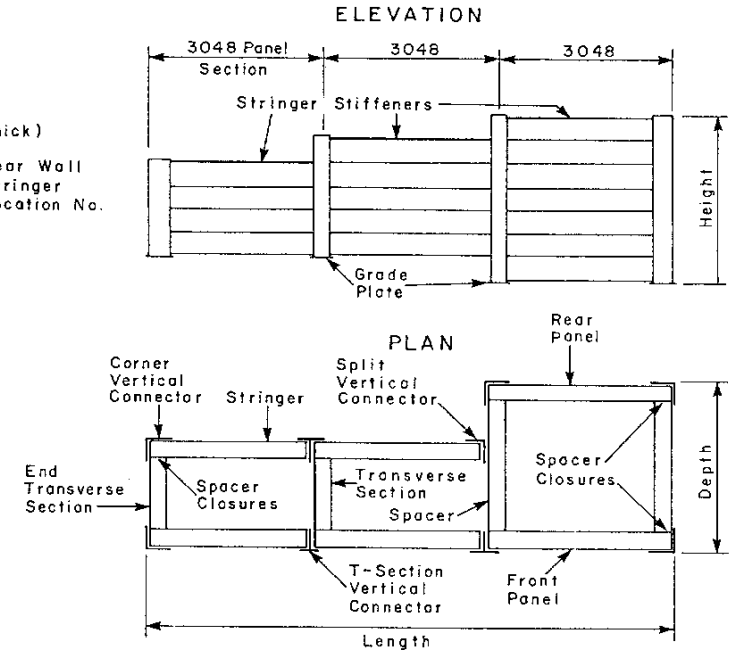
NOTES :
- Binwall Batter = 1:6 or Vertical as Required by Contract Drawings.
- Curvature Detail in accordance with Suppliers Manual.
- All measurements in millimetres.
- For Type 1 see Dwg. 3-SP 232.

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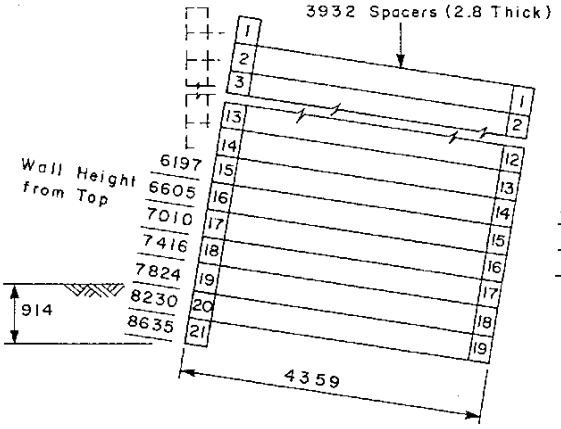
Revisions	TYPE 2 METAL BIN WALL DESIGNS A, B & C	
	Date 1985 11 06 — Approved M.G. Elston — Exec. Dir. of Engineering	SPECIFICATION DRAWING No. 1-SP 232

DESIGN D





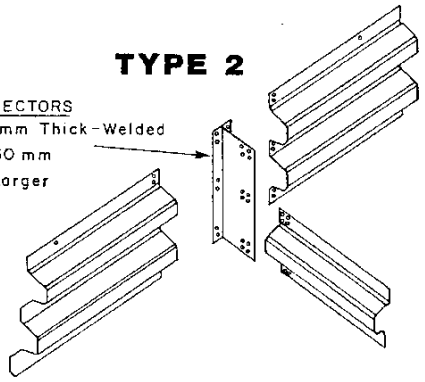
DESIGN E



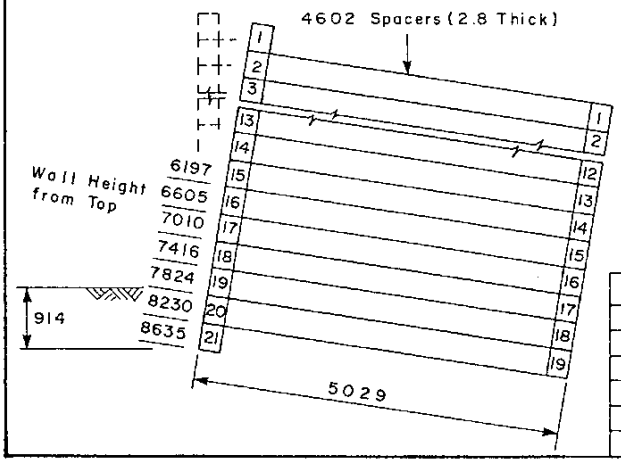
Front Wall Stringer Location No.	Rear Wall Stringer Location No.	Thickness
1 through 8	1 through 6	1.6
9 through 12	7 through 10	2.0
13 through 19	11 through 17	2.8
20 and 21	18 and 19	3.5

TYPE 2

VERTICAL CONNECTORS
- Steel Plate 6.3 mm Thick - Welded
- Max. Length 3660 mm
- Splice Plate if Larger



DESIGN F



NOTES :
- Binwall Batter = 1:6 or Vertical as Required by Contract Drawings.
- Curvature Detail in accordance with Suppliers Manual.
- All Measurements in Millimetres.
- For Type 1 see Dwg. 3-SP232.

Province of British Columbia Ministry of Transportation and Highways

Revisions	TYPE 2 METAL BIN WALL DESIGNS D, E & F	
	Date 1985 11 06 — Approved MGElston — Exec. Dir. of Engineering	SPECIFICATION DRAWING No. 2-SP 232